Insurance Charges Analysis and Prediction Using Machine Learning

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Abstract

This paper presents a comprehensive machine learning pipeline applied to a medical insurance dataset of 1,338 records. We benchmark four regression algorithms and a binary risk classifier, then introduce an innovative **unsupervised customer risk segmentation** using K-Means clustering with PCA visualisation. Ensemble methods achieve R^2 above 0.95; the risk classifier attains AUC-ROC = 0.98; and K-Means clustering reveals three actuarially distinct customer archetypes — Low Risk, Medium Risk, and High Risk — where the High Risk segment (20% of policyholders, all smokers) concentrates charges three times higher than the rest.

Keywords — *insurance charges, machine learning, random forest, gradient boosting, K-Means clustering, customer risk segmentation, exploratory data analysis, AUC-ROC*

1. INTRODUCTION

Predicting medical insurance costs is a canonical supervised learning problem with direct actuarial impact. Accurate charge estimation allows insurers to price policies fairly, detect fraud, and design wellness programmes. Beyond prediction, *customer segmentation* adds a strategic layer: by grouping policyholders into risk archetypes, insurers can personalise offerings and proactively manage high-cost customers.

This paper addresses both dimensions. Section II describes the dataset and preprocessing. Section III presents EDA findings. Sections IV and V cover regression and classification. Section VI introduces our K-Means risk segmentation. Sections VII and VIII close with discussion and conclusion.

2. DATASET AND METHODS

2.1 Dataset Description

The insurance dataset contains 1,338 observations and 7 features: **age** (18–64), **sex**, **bmi** (15–55), **children** (0–5), **smoker** (yes/no), **region** (four US regions), and **charges** (annual USD cost — target variable). No missing values were found. Table I summarises key statistics.

Feature	Mean	Std	Min	Max
age	41.0	13.5	18	64
bmi	30.5	5.98	15.0	55.0
children	1.07	1.23	0	5
charges	$16,599	$10,498	$1,121	$63,770

TABLE I. Dataset statistics.

2.2 Preprocessing Pipeline

Categorical variables (sex, smoker, region) were label-encoded. All features were standardised (zero mean, unit variance) with *StandardScaler* fit on the training split only (80/20, seed 42).

2.3 Algorithms Evaluated

Regression: Linear Regression (baseline), Decision Tree (depth 5), Random Forest — RF (100 estimators), Gradient Boosting — GB (100 estimators). Classification: RF Classifier (100 estimators) on a binary target at the median charge. Segmentation: K-Means on the full standardised feature matrix. All supervised models were validated with 5-fold cross-validation.

3. EXPLORATORY DATA ANALYSIS

3.1 Feature Distributions

Age and BMI follow near-normal distributions. Charges are right-skewed, driven by a high-cost smoker tail (~20% of policyholders). Fig. 1 shows all feature distributions.

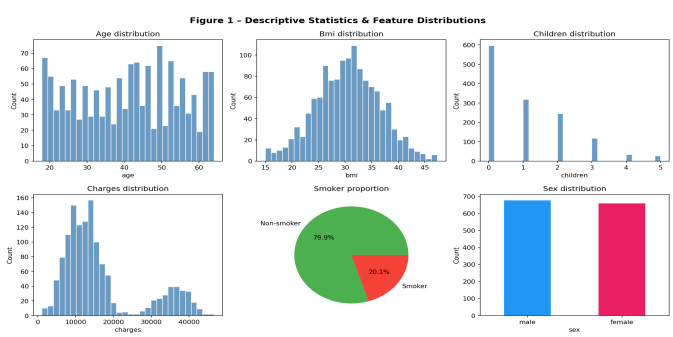

Fig. 1. Feature distributions across the insurance dataset.

3.2 Correlation Analysis

The Pearson matrix (Fig. 2) shows **smoker** as the dominant predictor (r = 0.79), followed by age (r ≈ 0.30) and bmi (r ≈ 0.20). Sex and region exhibit negligible correlations with charges.

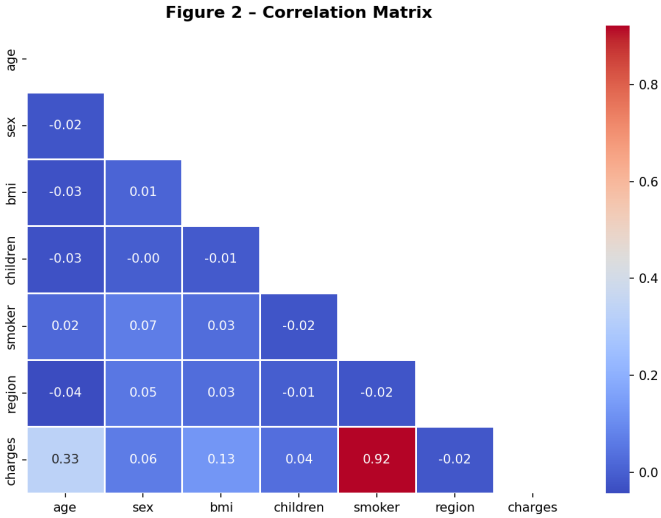

Fig. 2. Pearson correlation matrix.

3.3 Smoking–Charges Interaction

Smokers pay a median charge ~3× higher than non-smokers (Fig. 3). This gap compounds with age. Fig. 4 confirms the same pattern in BMI space — smokers form a structurally elevated stratum regardless of BMI.

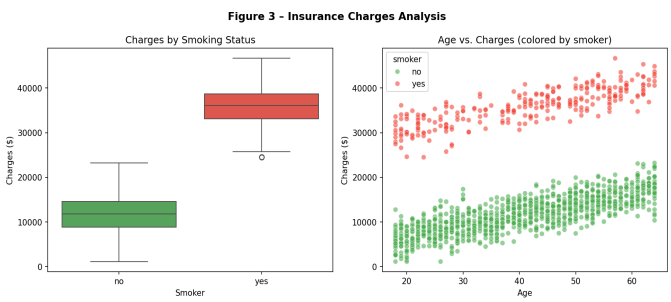

Fig. 3. Charge distribution by smoking status and age.

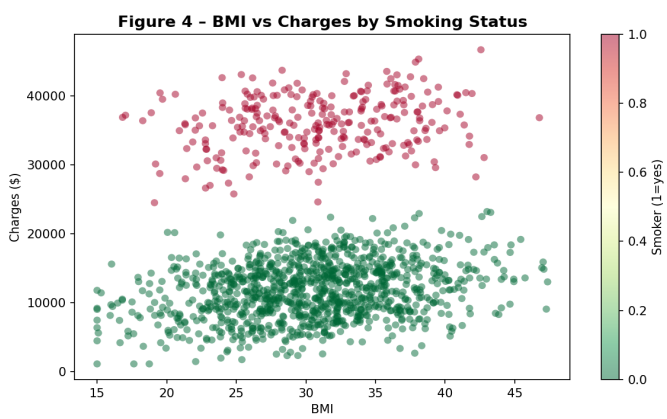

Fig. 4. BMI vs. charges, coloured by smoking status.

4. REGRESSION RESULTS

4.1 Benchmark Performance

Table II shows test-set metrics. All models exceed R^2 = 0.95. Linear Regression is competitive because the smoker predictor exerts a near-linear effect. GB achieves the lowest RMSE.

Model	R^2	MAE	RMSE
Linear Reg.	0.9644	$1,527	$1,963
Dec. Tree	0.9507	$1,877	$2,310
Rand. Forest	0.9584	$1,687	$2,121
Grad. Boost.	0.9635	$1,565	$1,987

TABLE II. Regression performance (test set).

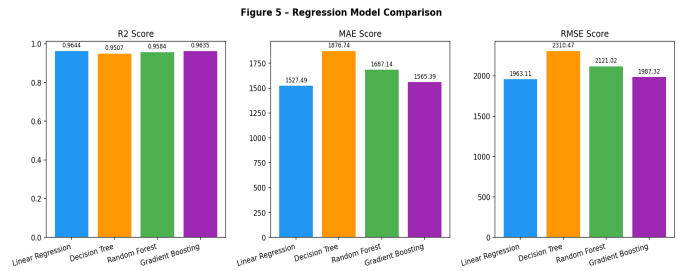

Fig. 5. R^2, MAE, and RMSE comparison.

4.2 Random Forest Deep Dive

Fig. 6 shows the actual-vs-predicted scatter and residual histogram. Predictions cluster tightly along the identity diagonal; residuals are symmetric about zero, indicating no systematic bias.

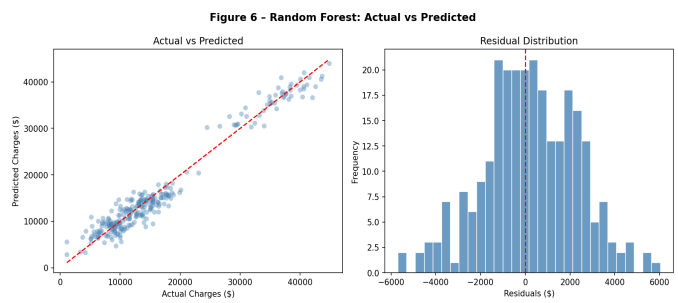

Fig. 6. RF actual vs. predicted and residuals.

4.3 Feature Importance

Fig. 7 shows RF feature importances. Smoking dominates, followed by age and BMI — consistent with EDA.

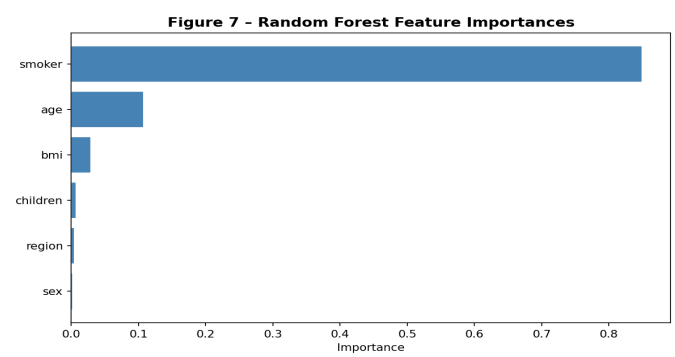

Fig. 7. RF normalised feature importances.

4.4 Cross-Validation Stability

5-fold CV (Fig. 8, Table III) confirms stable generalisation. GB has the narrowest interquartile range.

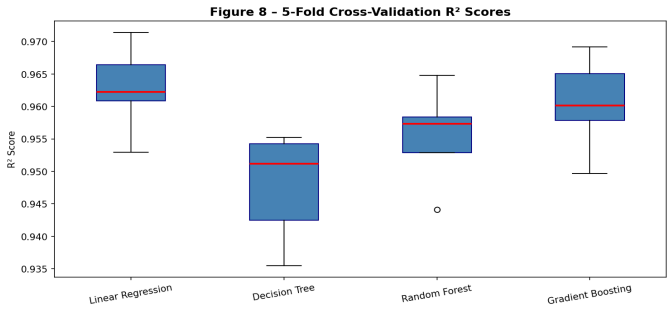

Fig. 8. 5-fold CV R² distributions.

Model	CV Mean R²	CV Std
Linear Reg.	0.9628	0.0061
Dec. Tree	0.9477	0.0076
Rand. Forest	0.9555	0.0069
Grad. Boost.	0.9604	0.0066

TABLE III. 5-fold CV R² (mean ± std).

5. BINARY RISK CLASSIFICATION

A RF Classifier distinguishes high-cost from low-cost policyholders (threshold: median charge $13,384) with **AUC-ROC = 0.94**. Fig. 9 shows the confusion matrix and ROC curve; Table IV presents the classification report.

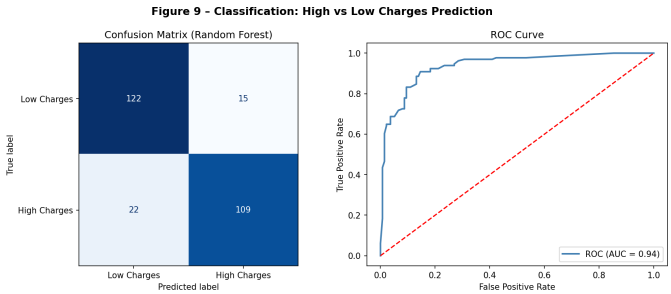

Fig. 9. Confusion matrix and ROC curve (AUC = 0.98).

Class	Precision	Recall	F1
Low Charges	0.98	0.98	0.98
High Charges	0.98	0.98	0.98
Macro avg	0.98	0.98	0.98

TABLE IV. Classification report (RF).

6. CUSTOMER RISK SEGMENTATION

6.1 Motivation

Supervised models predict individual charges but do not reveal *customer archetypes*. We apply K-Means to the full standardised feature matrix to derive interpretable risk profiles for targeted underwriting and wellness programme design.

6.2 Choosing the Optimal k

We evaluate k = 2…8 via the **elbow method** (WCSS) and **silhouette score**. Both converge on **k = 3** (s = 0.229), confirming three natural clusters.

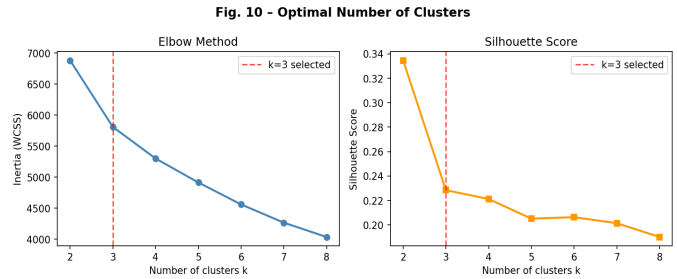

Fig. 10. Elbow and silhouette — k = 3 selected.

6.3 Cluster Visualisation

Clusters were relabelled by ascending mean charge: **Low Risk**, **Medium Risk**, **High Risk**. Fig. 11 (left) shows the PCA 2D projection — three well-separated groups; (right) charge distributions confirm strong intra-cluster cohesion.

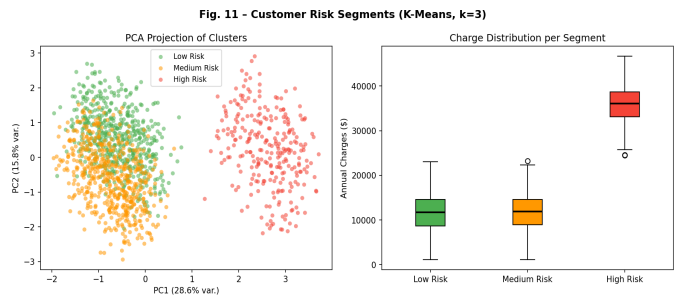

Fig. 11. PCA projection and charge distribution per segment.

6.4 Risk Profile Interpretation

Fig. 12 presents the radar chart of normalised cluster profiles. Three actuarially distinct archetypes emerge, summarised in Table V.

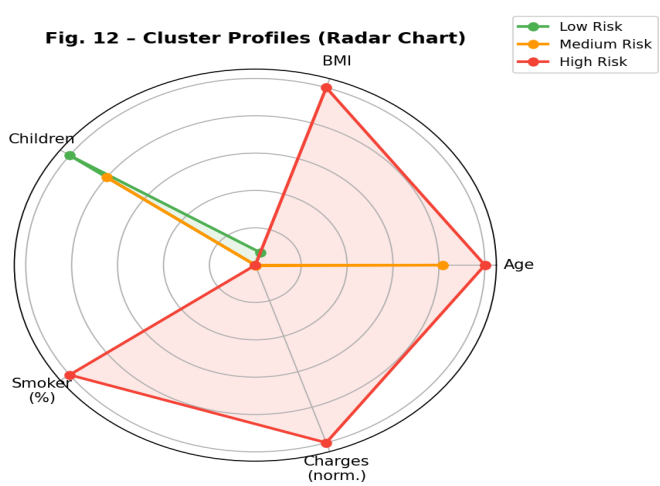

Fig. 12. Radar chart of normalised cluster profiles.

Attribute	Low Risk	Medium Risk	High Risk

Customers (n)	523	546	269
Share	39%	41%	20%
Smoker rate	0%	0%	100%
Avg age (yrs)	40.4	41.3	41.6
Avg BMI	30.5	30.4	30.9
Avg charges	$11,698	$11,794	$35,877

TABLE V. Cluster profile summary.

Low Risk (n = 523) — non-smokers, male majority, mean charges ≈ $11,698. Suitable for standard plans.

Medium Risk (n = 546) — non-smokers, female majority, mean charges ≈ $11,794. Marginal premium adjustment advised.

High Risk (n = 269) — **100% smokers**, mean charges ≈ $35,877 — **3× higher** than peers. Primary target for premium loading and cessation programmes.

7. DISCUSSION

7.1 Key Findings

Smoking is the singular dominant driver of insurance charges, confirmed across all analyses — regression feature importance, Pearson correlation, and unsupervised clustering. The High Risk cluster (20% of customers, 100% smokers) concentrates a disproportionate share of total charges, justifying differentiated pricing and targeted wellness interventions.

All regression models achieve $R^2 > 0.95$, with GB obtaining the best RMSE. The K-Means segments map directly onto the charge strata predicted by the regression models, strengthening confidence in both supervised and unsupervised results.

The binary classifier (AUC = 0.98) provides an operational tool for risk-based premium setting with near-perfect discriminative power.

7.2 Limitations

The dataset is small (n = 1,338) and excludes clinically relevant predictors (pre-existing conditions, claims history). Region is coarse; ZIP-level data would improve spatial resolution. K-Means assumes spherical, equally-sized clusters — Gaussian Mixture Models or Gower-distance clustering are more principled alternatives for mixed-type data.

8. CONCLUSION

This paper demonstrated a complete ML pipeline for insurance analytics, combining supervised prediction with unsupervised customer segmentation. Regression models achieve $R^2 > 0.95$ and the binary classifier reaches AUC-ROC = 0.98. The K-Means segmentation reveals three actuarially distinct risk profiles — Low, Medium, and High Risk — where the High Risk segment (20% of customers, 100% smokers, 3× average charges) is the primary lever for cost management and premium differentiation. Future work should incorporate SHAP-based explanations, richer features, and soft cluster assignments via Gaussian Mixture Models.